DATASCI 530 – Project 3

For the final project, you are expected to produce a report and a final presentation at the end of class. You have several options for choosing a dataset. You only need to choose one option.

Option 1: Scraped dataset

Expand the web scraping analysis from the midsemester project.

(1) Extract product-specific text (e.g., product descriptions, reviews, etc.), which is often available after clicking on each product. If you already collected this information for Projects 1 and 2, you do not need to do this again.
(2) You are expected to apply an unsupervised machine learning algorithm for text analysis. This can include topic modeling, clustering, or other text analysis methods.

Option 2: Third-party data with individual text records. It should contain at least 2,000 records.

(1) You are expected to apply an unsupervised machine learning algorithm for text analysis. This can include topic modeling, clustering, or other text analysis methods.

Option 3: Third-party data with images. It should contain at least 2,000 records.

(1) Apply a supervised machine learning algorithm.

For all three options, there should be a strong focus on modeling. You should use appropriate model evaluation metrics, conduct appropriate hyperparameter tuning, and clearly justify your modeling choices. Your project should be written in the form of a report in which you describe the dataset, its source, your methodology, and the major insights.

FINAL PRESENTATION:

After you have submitted your Project 3 deliverable, you will be expected to present your findings to your classmates during the "In-Class Workshop" (check the syllabus for scheduled dates). Prepare a 5-7 minute presentation with slides. The slides do not need to be submitted in advance. There will be a question-and-answer (Q&A) session after your presentation.

Rubric:

You will be graded both on the code and the writing using the following point system.

The use of generative AI is encouraged for diagnosing problems, debugging code, and learning how to use new packages. However, AI use should be focused on specific tasks, and AI-generated code should be reviewed and edited to suit the needs of the project. The final

product should be polished, concise, and readable. Excessively long or unedited AI-generated code may receive fewer points.

Component	Detailed Points	Total Points
Overall		3
Organization and documentation of code.	2	
Originality.	1	
Introduction		2
(a) Description of topic and question. The report should provide context about the data and what we learn from it.	1	
(b) Summary of findings.	1	
Data Analysis		8
(a) Explanation of methods used and why they are suitable for the analysis.	2	
(b) The methods are implemented in a conceptually correct way (e.g., appropriate use of cross-validation without data leakage, hyperparameter tuning, and performance metrics).	3	
(c) There is a clearly defined research question, and the discussion explains how the findings relate to it. The report includes sufficient diagnostic metrics that allow the reader to understand the strengths and limitations of the approach.	3	
Results		6
(a) Quality checks.	2	
(b) Table presenting key information. Tables are properly formatted (e.g., have appropriately rounded values, relabeled columns, captions, and are not too long).	2	
(c) Figures presenting key information. There should be a strong emphasis on interpreting the patterns and insights revealed by the data. All graphs look professional (have axis and title labels, a suitable color scheme, appropriate font sizes, a legend when needed, etc.).	2	
Discussion		
Clarity and conciseness.	1	1
Total		20

Your final presentation will be graded separately than the Project 3 deliverable.